



Opportunity Validation · Juthoor (جذور)

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1. Problem, Customer & User

Problem. In prerequisite-based subjects, a student can fail a current lesson because an earlier concept is weak. The student sees the wrong answer; the parent sees the mark; neither necessarily sees where the gap began. Juthoor traces evidence backward through prerequisite links, identifies the most likely root only when the evidence threshold is met, and sends the student back to that foundation before returning to the current lesson.

Role	Current definition
User	Student; initial audience Grades 5–7, with Grade 6 mathematics as the current MVP scope.
Economic buyer	Parent/guardian. Free 0 JOD; Pro Monthly 4.50 JOD/month; Pro Academic Year 32 JOD/year.
Channels / validators	Teachers and schools may recommend Juthoor or review diagnoses. They are not paying customers in the current B2C model.

Scope discipline. Juthoor is designed as a prerequisite-aware adaptive learning and root-gap diagnosis platform, not a mathematics-only product. The current MVP and all current evidence are mathematics: nine Grade 6 lessons covering integers and fractions. Other subjects and grades are planned validation, not existing evidence.

2. Evidence of the Problem

Jordan had 2,307,110 school students in 2023/24, including 583,685 in private schools [1]. Jordan's PISA 2025 mathematics score was 388 versus an OECD average of 463 [2]. A dated Jordan tutoring study reported that 54% of families used private tutoring and mathematics was the most-tutored subject (33.3%), with reported family spending of roughly 100–150 JOD/month [3]. These sources provide context and existing spending around learning support; they do not prove demand for Juthoor.

Our anonymous student questionnaire (34 responses; convenience sample) found 21/34 often or always cited forgotten basics, 30/34 could not pinpoint the old gap alone, and 33/34 rated a step-by-step return important. These are student perceptions, not parent or purchase evidence.

3. Three Key Assumptions

Assumption	Evidence we have	Not yet validated
A1 — The problem is real for students.	21/34 often or always cite forgotten basics; 30/34 cannot find the gap alone; 33/34 value step-by-step return.	Parent interviews; observed product behavior; representative sample.
A2 — Diagnosis is directionally useful.	Six anonymised mathematics cases: 1 Yes, 5 Partial, 0 recorded disagreements against one educator's recorded root.	Verified ground truth; blind review; multiple educators; larger sample; clear Partial criteria.
A3 — Parents will pay and students will return.	Student proxy only: 16 yes, 6 maybe, 12 no to a symbolic price. Price was not stated and students are not buyers.	Parent WTP, conversion, annual-plan choice, retention, referral.

Methodological limit. The six-case record has no documented provenance, dates, setting, educator identity, or blind-condition evidence. We therefore treat it as an early signal, not a diagnostic-accuracy study.

Next validation round: test a larger case set with at least two independent educators reviewing roots before seeing Juthoor's output, and report exact agreement, partial agreement, disagreement, and abstention rates.

4. Six-Case Signal

Case	Difficulty	Educator root	Juthoor root/focus	Agreement	Pre→post
S01	Integers & absolute value	Weak negative numbers	Number line; sign vs distance	Partial	4→9
S02	Adding integers	Adding negative numbers	Movement/direction	Partial	3→8
S03	Subtracting integers	Number sign vs operation sign	Sign vs operation; inverse	Yes	5→9
S04	Multiplying integers	Sign rules	Calculation vs sign; fact gap	Partial	4→8
S05	Adding/subtracting fractions	Common denominators	Equivalent fractions/value	Partial	3→9
S06	Adding mixed numbers	Adding fractions	Whole + fraction/value	Partial	4→8

The six scores average 3.83/10 before and 8.50/10 after, a +4.67-point observed change. There is no control group, and the protocol does not document the test forms, timing or administration conditions for these six scores; no causal learning claim is made.

The separate synthetic engine benchmark reported 97.1% correct among committed diagnoses and 89.8% overall. This tests logic on simulated learners, not real-student diagnostic accuracy.

5. What We Learned & Changed

Finding	Decision	Evidence status
Students report difficulty locating earlier gaps.	Keep root-gap diagnosis at the center.	Student-perception signal only.
Six selected cases show topic-level alignment but only one exact match.	Keep evidence-first diagnosis and expand independent validation.	Early signal; accuracy unproven.
No school/teacher commitment and no parent purchase data.	Use B2C: parent pays, student uses.	Parent demand is the next test.
Wrong diagnosis can destroy trust.	Require sufficient evidence; otherwise “insufficient evidence”; show ordinal confidence, not fake probability.	Implemented; calibration unproven.

6. Final Value Proposition

For students whose current lesson depends on earlier concepts, and for parents who see a weak result without knowing why, Juthoor is a **prerequisite-aware adaptive learning platform** that finds the **most likely earlier gap** behind a student's errors, shows the **evidence and confidence label**, and **guides the student back to that foundation** before returning to the current lesson.

Unlike a generic chatbot or content library that mainly answers the question in front of the learner, Juthoor focuses on the **prerequisite chain**. A **free diagnostic lets a family try it first**; **Pro adds the full root-chain diagnosis and printable parent gap report**. The current price is a testable hypothesis: **4.50 JOD/month or 32 JOD/academic year**.

Sources

- Department of Statistics, Jordan Statistical Yearbook 2024 – Education.
- OECD Education GPS – Jordan, PISA 2025.
- “Private Tutoring in Jordan: Underpinning Factors and Impacts”, IJHSS 3(13), 2013.
- Al-Adaa Center, Assessment of the Current State of Education in Jordan (2023/2024).
- TechSparks Juthoor MVP repository: docs/PILOT.md and app/engine/diagnosis.py.
- TechSparks six anonymised cases S01–S06; provenance not documented.
- TechSparks student questionnaire, 34 responses, 1–3 October 2026; convenience sample.
<https://docs.google.com/spreadsheets/d/1W6Tno15wQeX0cB04X5uOe6JS1INLi1bD/edit?usp=sharing&ouid=113409267831519981002&rtopof=true&sd=true>
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